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Noninvasive detection of diabetes in obstructive sleep apnea based on overnight SpO_2 signal and deep learning

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***Abstract*—The prevalence of obstructive sleep apnea comorbid with diabetes is high while the awareness of diabetes is low. There is a strong need for new diagnostic biomarkers to detect diabetes at an early stage. Therefore, we aimed to establish an automatic, deep-learning based model that could be applied to assess diabetes risks using overnight SpO_2 signals. The samples were derived from the Sleep Heart Health Study including 5,021 middle-aged and older adults (6.9% diabetes). The deep-learning models were established to identify diabetes solely from SpO_2 or in combination with clinical factors (gender, age, and BMI). Class Activation Map (CAM) was utilized to determine the models' effects. By adding SpO_2 to clinical factors, the prediction performance was significantly improved from 0.646 ± 0.011 to 0.751 ± 0.006 in terms of the area under the receiver operator characteristic (AUC) after 10-fold cross-validation. CAM results showed significant subsequences for the classification decision were the hypoxic events. The findings suggest that the SpO_2 signals could provide substantial information. The deep learning model could be used to evaluate diabetes risks, which is beneficial for long-term health management monitoring.**

Keywords—diabetes, obstructive sleep apnea, oxygen saturation, deep learning, class activation map

I. Introduction

Type 2 diabetes is a major chronic disease, with an estimated prevalence of 9.3% in 2019 [1]. It is also a leading risk factor for many other common medical problems, including cardiovascular disease, kidney disease and retinal disease [2]. The disease progression is hidden, and it needs intensive monitoring of blood glucose and other indicators to make a precise diagnosis, leading to late diagnosis and low awareness. It is estimated that up to 193 million people have undiagnosed diabetes around the world [3]. Therefore, diabetes screening and diagnosis at an early stage are critical.

Obstructive sleep apnea (OSA) is a common sleep disorder characterized by recurrent episodes of complete (apnea) or partial (hypopnea) obstruction of the upper airway and frequent comorbidity in patients with type 2 diabetes [4]. The prevalence of type 2 diabetes was threefold higher in people with OSA than those without OSA, according to the Sleep Heart Health Study [5]. Intermittent hypoxemia (IH) and sleep fragmentation are cardinal features of OSA and are likely in the causal pathway leading to metabolic dysfunction [6, 7]. Therefore, we hypothesized the nocturnal oxygen saturation (SpO_2) of OSA patients could be applied to predict the risk of diabetes to a certain extent.

With the development of artificial intelligence, information that was not reflected in the traditional manual features of SpO_2 signals can be effectively extracted directly through deep learning to figure out the relationship between intermittent hypoxia and diabetes. In addition, it is relatively easy to obtain the SpO_2 signals, which can be collected with wearable devices such as smartwatches and rings in a home-based setting. Effective analysis is conducive to improving the health management of OSA patients.

To date, effective diagnosis models for diabetes in OSA have not been found. Therefore, this study aimed to develop an automatic, deep-learning based model without manual feature extraction to make noninvasive and outstanding detection of diabetes for individuals. The model took one night of SpO_2 signals as input. The Class Activation Map (CAM) was utilized to figure out the effect of the model and explain the recognition ability of the network.

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II. Materials and Methods

This study was approved by Beijing Tsinghua Changgung Hospital Ethics Committee Ethics (No.22561-6-01). All methods were conducted in accordance with relevant guidelines (Declaration of Helsinki) and regulations.

A. Data Source

This cross-sectional study was based on the Sleep Heart Health Study (SHHS), a community-based study designed to explore the associations between sleep-disordered breathing and related health outcomes. This sample included 6,441 men and women aged 40 years or older who completed a baseline unattended type 2 polysomnogram (PSG) (1995-1998) and a standardized questionnaire. Individuals treated with CPAP, oral devices, nocturnal supplemental oxygen, or tracheostomy were excluded [8]. The data use agreement for the current study was obtained from the National Sleep Research Resource. Local institutional review board approval was not sought for this study given that the terms of use of the deidentified data were governed by the National Sleep Research Resource. Patient consent was not obtained given that no identifiable information was used and that all fields were anonymized at source. The data set was downloaded and analyzed in September 2021. More detailed information and design have been reported previously [8].

Of the 6,441 PSGs, 5,804 studies were available through National Sleep Research Resource (https://sleepdata.org/). After excluding 610 cases with respiratory failure, 111 with sleep apnea treatment, and 62 without PSG or with poor SpO_2 quality, a total of 5021 individuals were available for analysis (Fig. 1). Diabetes was defined based on self-report or usage of oral hypoglycemic agents or insulin at enrollment, as in previous SHHS publications [9]. Since there were 4674 non-diabetes and 347 diabetes, we selected at random to match the number of diabetes to address the data imbalance, which was also applied in other work [10].

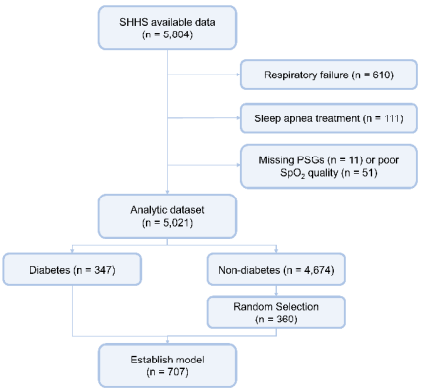


Fig. 1. Flow diagram presenting ascertainment of the study samples. PSG, polysomnogram; SHHS, the Sleep Heart Health Study; SpO_2, oxygen saturation.

B. PSG and SpO_2 Parameters

SpO_2 signals were measured by fingertip pulse oximetry (Nonin, Minneapolis, MN) sampled at 1 Hz. We removed the outliers of <10 and filled in missing values with linear interpolation. To make the SpO_2 signal as input the same length, we used zero padding to 8 hours. The apnea-hypopnea index (AHI) was calculated using apneas and hypopneas both associated with desaturations of ≥ 4%. Other PSG variables included oxygen desaturation index (ODI), Minimum SpO_2, Average SpO_2, and percent time below oxygen desaturation of 90% (CT90).

C. Neural Network Architecture

The overall architecture of the proposed model is depicted in Fig. 2. The input was SpO_2 signal and clinical variables including gender, age, body mass index (BMI), and the output was diabetes risk values. The network training was performed using categorical cross-entropy loss function and an Adam optimizer with a learning rate of 0.01 and weight decay of 0.01 for 500 iterations by Python 3.7.7 with Pytorch 1.3.1. Finally, the use of CAM was investigated to evaluate the effects of the proposed model and explain the recognition ability of a network.

D. Statistical Analysis

All statistical analysis was performed with IBM SPSS Statistics version 26. The concentrating and dispersing trends of the data were described by X ± SD. A Mann–Whitney U test was used for comparisons between groups and categorical data were analyzed by the Chi-square test. A p value$<$0.05 was considered significant, and all p values were two-sided.

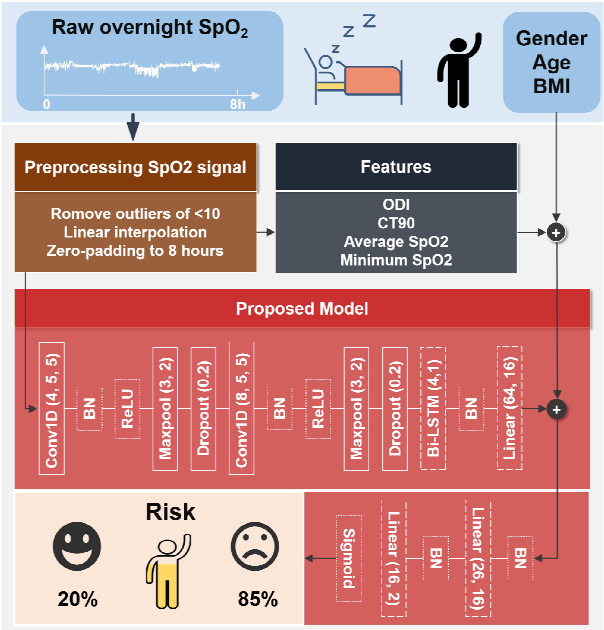


Fig. 2. Illustration of the architecture of the combined convolutional neural network (CNN) and long-short term memory (LSTM). The CNN comprised three 1D convolutions (Conv1D), batch normalizations (BN), rectified linear unit (ReLU) activation functions, and max-poolings. The parameters of the convolutional layers are given as (number of filters, kernel size, stride size) and the parameters of the max-pooling layers are given as (pool size, stride size). The parameters of the LSTM are given as (hidden size, layers) The dropout rate is given for the dropouts. The CNN was followed by a bidirectional LSTM, and linear layers with a sigmoid activation function.

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III. Results

A. Baseline characteristics

Table I shows the demographics and sleep characteristics of the subjects grouped by diabetes status. 6.9% of participants (n = 5,021) were diagnosed with diabetes, while 10.2% of participants with severe OSA based on an AHI ≥ 30 events/hour (n = 767) were diagnosed with diabetes. Subjects with diabetes were older, more obese, and had more severe OSA than those without diabetes. There were no significant differences between the random selection dataset and the primary non-diabetes dataset in terms of demographics and sleep characteristics (Table II).

TABLE I. Characteristics by Diabetes Status

	Diabetes N = 347	Non-diabetes N = 4674	*P*-value
Male	201 (57.9%)	2212 (47.3%)	<0.001
Age (years)	69±10	63±11	<0.001
BMI (kg/m^2)	30.1±5.5	27.9±4.9	<0.001
AHI (events/hour)	21.4±16.2	16.6±14.5	<0.001
Hypertension	264 (76.1%)	2167 (46.4%)	<0.001
ODI (events/hour)	3.8±7.5	2.9±6.6	0.001
Average SpO_2 (%)	94.2±2.2	94.6±2.0	<0.001
Minimum SpO_2 (%)	84.6±5.9	85.9±6.1	<0.001
CT90 (%)	5.4±14.9	3.0±9.3	<0.001
ESS	8.5±4.8	7.6±4.3	0.001

Parameter values are presented as means ± standard deviations or number (percent). AHI, apnea hypopnea index (apneas and hypopneas associated with ≥ 4% desaturation); ODI, oxygen desaturation index; CT90, percent time below oxygen desaturation of 90%; ESS, Epworth Sleepiness Scale.

TABLE II. Sample Characteristics

	Non-diabetes N = 4674	Random selection from non-diabetes N = 360	P value
Male	(47.3%)	(42.8%)	0.096
Age (years)	63±11	63±11	0.618
BMI (kg/m^2)	27.9±4.9	27.6±4.5	0.561
Hypertension	2017 (43.2%)	150 (41.7%)	0.069
AHI (events/hour)	16.6±14.5	16.6±14.4	0.957
ODI (events/hour)	2.9±6.6	3.0±5.9	0.905
AverageSpO_2 (%)	94.6±2.0	94.8±1.9	0.289
MinSpO_2 (%)	85.9±6.1	85.8±6.1	0.954
CT90 (%)	3.0±9.3	2.8±8.6	0.700
ESS	7.6±4.3	7.4±5.0	0.166

B. Model performance

We carried out 10-fold cross-validation and used measures including accuracy, sensitivity, specificity, precision, F1-score, and area under the curve (AUC) to estimate the proposed model.

By adding SpO_2 to clinical variables, the deep learning model achieved a classification accuracy of 69.9%, an AUC of 75.1%, a sensitivity of 73.1%, a precision of 67.1%, and an F1-score of 69.5%, all of which were higher than those in models with SpO_2 or variables alone. (Table III).

TABLE III. Performance with Different Input

TABLE IV. Performance	Variables	SpO_2	SpO_2 and variables
AUC	0.646±0.011	0.601±0.018	**0.751**±0.006
Accuracy	0.668±0.019	0.617±0.023	**0.699**±0.013
Sensitivity	0.689±0.027	0.628±0.028	**0.731**±0.029
Specificity	**0.647**±0.014	0.586±0.034	0.633±0.024
Precision	0.647±0.014	0.587±0.017	**0.671**±0.004
F1-score	0.657±0.019	0.592±0.019	**0.695**±0.012

Variables included gender, age, BMI, ODI, average SpO_2, minimum SpO_2 and CT90.

C. Visualization

We investigated the use of a one-dimensional CAM proposed by Wang et al. [11] for visualization to find out the subsequences of time series which contribute the most to classification. The CAM results of the model are shown in Fig. 3. The discriminative regions of the time series highlighted to show significant subsequences for the classification decision were the hypoxic events.

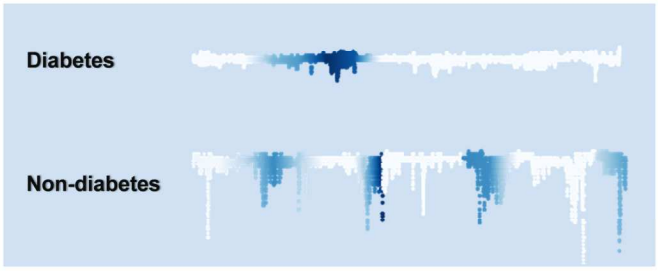


Fig. 3. CAM results. The discriminative regions of time series were highlighted to show important subsequences for the classification decision, and the faded region showed no contribution to the classification.

IV. Discussion

In the present study, we established a deep-learning based model to enhance the diagnostic ability of the overnight SpO_2 signal and clinical variables. The proposed model showed a high diagnostic ability, improving the diagnostic performance of clinical variables such as age, BMI, and ODI. The model achieved a classification accuracy of 69.9%, an AUC of 75.1% and a sensitivity of 73.1% in a large database. In a previous study, Ozal Yildirim developed an automated detection of diabetes using heart rate signals based on deep learning, which achieved an average accuracy of 97.6% and a sensitivity of 100%. However, this study was limited as the sample size was only 30 [12].

Application of the SpO_2-based model could assist in detecting diabetes in multiple consumer-grade health technology devices. Nowadays, reflective SpO_2 sensors are integrated into various wearable self-tracking devices, such as activity wristbands and smartwatches. SpO_2 monitoring collects multiple nighttime data, making the detection more reliable. Moreover, when compared with blood test, SpO_2 monitoring is noninvasive and considerably cheaper to conduct. The majority of diabetes patients remain undiagnosed with estimates of unrecorded cases of at least 30% in Mexico City and 26% in South Asia [13, 14]. The application of the

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SpO_2 based model could raise people's awareness of their symptoms of diabetes, although some of them do not consider themselves as patients.

Multiple mechanistic pathways are likely to be causally involved between OSA and diabetes, including increased sympathetic neural activity, oxidative stress, systemic inflammation, activation of HPA axis and alteration of circulating adipokines [6]. Recent studies, concerning deriving landmark-based, hand-selected features, have identified that intensity, duration, and depth of IH are closely associated with diabetes. Gonzalo Labarca reported that CT90 ≥ 20%, ODI ≥ 30 events/hour, and minimum SpO_2 < 70% were linked to an increased risk of diabetes [15]. Consistent with this study, Kendzerska also discovered that CT90 was associated with incident diabetes [16].

In the proposed model, the multi-scale convolution captured different scales of information and LSTM took account of the short-term, and long-term information to achieve remarkable performance. However, a deep learning model is known as a black box, which is hard to explain the output. In this study, by up-sampling CAM to the size of the SpO_2 signal, the most essential subsequences can be identified. It is noted that the discriminative regions focused on the hypoxic events, which suggested that the features of hypoxic events may be closely related to diabetes. Many IH parameters are associated with OSA-related cardiovascular comorbidities. Nana Wang proposed that the oxygen desaturation rate in severe OSA was strongly associated with hypertension [17]. Whether the IH parameters are related to diabetes remains unclear and needs to be studied.

We considered several limitations of the current study. First, the cross-sectional study cannot reveal the causal relationship between IH and diabetes. A long-term effect of nocturnal IH on the risk of type 2 diabetes will be explored in the future. Second, SpO_2 signals during one night might not provide enough information to detect diabetes. Although there is no significant difference between mean AHI in two sequential study nights [18], additional studies are needed to validate the method during multiple nights.

V. Conclusion

This study proposed a novel, deep-learning based method for automated and noninvasive diagnosis of diabetes using SpO_2 signals and clinical variables. The deep learning approach could be easily applied to wearable devices to keep long-term monitoring of health management.

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